

Surveillance Regime Model

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1 Model Setup

In what follows, I present a formal model that clarifies one part of Chapter 2’s broader theory: how regimes combine surveillance and visible repression when controlling dissident threats. The model does not reproduce the full dynamic feedback loop between surveillance, repression, concealment, and further surveillance expansion. Instead, it formalizes the static identification–deterrence tradeoff: surveillance helps rulers identify hidden opponents, while repression deters dissent but can also induce concealment and weaken surveillance’s informational value. This tradeoff explains why surveillance and visible repression may sometimes substitute for one another and sometimes complements one another—the distinct logics behind surveillance authoritarianism and punitive surveillance states. It also clarifies why digital surveillance can sustain surveillance authoritarianism by lowering the social and legitimacy costs of surveillance, reducing fear, and enabling cheaper targeted enforcement.

1.1 Player and Strategies

In this one-player model, a regime choose a level of surveillance s and a violent level of repression r to reduce threats imposed by dissidents in society. r indicates the degree of overt violence observable to citizens, with higher values corresponding to more intensive repressive measures. Citizens and dissidents are non-strategic: their responses are incorporated in the payoff function of the regime. Thus, the framework constitutes an optimization problem faced solely by the regime.

1.2 Identification, Deterrence, & Residual Threat

A fraction α of citizens are potential dissenters. $\alpha \in [0, 1]$. The regime has information about this fraction α but does not know who these citizens are before it invests in surveillance.

Under technology tech , surveillance level s , and repression level r , the regime’s *identification effectiveness* is

$$P^{(\text{tech})}(s, r), \quad \frac{\partial P^{(\text{tech})}}{\partial s} > 0, \quad \frac{\partial P^{(\text{tech})}}{\partial r} < 0.$$

P captures the the capability of successfully identifying a dissident given that repression reduces the identification effect of surveillance.

Likewise, the regime's *deterrence effectiveness* is

$$Q^{(\text{tech})}(s, r), \quad \frac{\partial Q^{(\text{tech})}}{\partial s} > 0, \quad \frac{\partial Q^{(\text{tech})}}{\partial r} > 0.$$

Q captures the deterrent effects of repression and surveillance. Note that P and Q are indices instead of probabilities; higher values indicate stronger detection or deterrence.

The *residual threat* falls with overall effectiveness via a separable kernel

$$\alpha F[P^{(\text{tech})}(s, r), Q^{(\text{tech})}(s, r)], \quad \frac{\partial F}{\partial P} < 0, \quad \frac{\partial F}{\partial Q} < 0.$$

Let's assume a convenient separable specification:

$$F = \exp\{-[P^{(\text{tech})} + Q^{(\text{tech})}]\},$$

which induces multiplicative reductions in threat and preserves tractability. The model is also tractable with linear function forms, which generate similar results.

Surveillance technology is exogenously given: $\text{tech} \in \{D, I\}$ (Digital, in-person). Assume $Q^{(D)}(s, r) < Q^{(I)}(s, r) \quad \forall s, r$, which captures the fact that in-person surveillance often relies on intimidation to function and has a stronger chilling effect on other dissidents.

Assume $P^{(D)}(s, r) = P^{(I)}(s, r) \quad \forall s, r$. This is reasonable because, although in-person surveillance combined with repression has a stronger chilling effect than digital surveillance, the associated intimidation was systematically leveraged to compel informer compliance and thereby facilitate the collection of additional information for the effectiveness of identification.

1.3 Costs of Surveillance and Repression

The regime incurs surveillance costs, $c_s^{(\text{tech})}(s)$, and costs of overt repression, $c_o(r)$. It also incurs costs from targeted enforcement, $c_t(P)$, which depend on the effective identification of dissidents, P . The total cost of targeted enforcement can be represented by the workload term

$$c_t(P) = \alpha c_t \exp\{\gamma P^{\text{tech}}(s, r)\}.$$

This workload term mechanically varies with P , while c_t is a constant that is independent of r , which helps preserve the tractability of the model.

Separating the cost of targeted enforcement from the regime's display of violence is reasonable because the theory focuses on repression that is visible to citizens, whereas the number of dissidents subjected to targeted enforcement need not determine the level of visible repression. For example, the regime could violently repress a single dissident and publicize the act in order to deter others. More generally, the regime can choose how much violence to reveal while using various non-public methods to deal with identified dissidents.

The parameter c_t can be interpreted as the "technology" of targeted enforcement: a higher value means that targeted enforcement is more costly for the regime. This parameter is taken as exogenous in the model. For example, imprisoning a dissident may be more costly than placing that individual on a blacklist or banning them from traveling to the capital.

$c_s^{(\text{tech})}(s)$ captures the costs of surveillance under different technologies. Assume

$$c_s^{(\text{D})}(s) < c_s^{(\text{I})}(s),$$

meaning that digital surveillance is less costly than in-person surveillance at a given level of surveillance. This assumption captures the legitimacy costs of surveillance in addition to its direct investment costs. As discussed above, in-person surveillance often faces stronger legitimacy costs and public resistance because of its intrusiveness and visibly repressive function. By contrast, digital surveillance is less visibly intrusive, and its public-good function helps justify its use and obscure its repressive function.

The direct investment costs of these two types of surveillance are likely to be broadly comparable. In-person surveillance incurs costs primarily through the recruitment, coordination, and maintenance of agents and informants, whereas digital surveillance requires substantial upfront investment in infrastructure, such as camera networks, data centers, fiber-optic lines, and other transmission systems. Although digital surveillance may have lower marginal monitoring costs once installed, its large fixed costs can make its overall investment cost comparable to that of in-person surveillance over the relevant range of s . Nevertheless, digital surveillance also provides public goods and can therefore be partly funded through public resources. This dual functionality may make digital surveillance less costly than in-person surveillance even in terms of investment costs.

1.4 The Regime's Objective Function

Let $\bar{V}$ be the baseline hold-power.

$$\begin{aligned} u^{(\text{tech})}(s, r) = & \bar{V} - \alpha F[P^{(\text{tech})}(s, r), Q^{(\text{tech})}(s, r)] \\ & - c_s^{(\text{tech})}(s) - \alpha c_t \exp\{\gamma P^{\text{tech}}(s, r)\} - \phi c_o(r). \end{aligned}$$

For a tractable objective function, let's assume quadratic cost functions for r and s . Let

- $P^{(\text{tech})}(s, r) = \beta_s s - \beta_r r + b_P,$
- $Q^{(\text{tech})}(s, r) = \delta_s s + \delta_r r + \tau_{\text{tech}} + b_Q,$

with $\beta_s, \beta_r, \delta_s, \delta_r > 0$ and $\tau_I > \tau_D$.

The objective function simplifies to:

$$\begin{aligned} u^{(\text{tech})}(s, r) = & \bar{V} - \alpha \exp(-[P^{(\text{tech})}(s, r) + Q^{(\text{tech})}(s, r)]) \\ & - \frac{1}{2} k_s^{(\text{tech})} s^2 - \alpha c_t \exp\{\gamma P^{\text{tech}}(s, r)\} - \frac{1}{2} \phi k_o r^2 \end{aligned}$$

2 Reduced Form Solutions

2.1 First-order conditions

For a fixed technology tech , define

$$A_s \equiv \beta_s + \delta_s > 0, \quad A_r \equiv \delta_r - \beta_r,$$

and

$$B_{\text{tech}} \equiv b_P + b_Q + \tau_{\text{tech}}.$$

Then

$$\begin{aligned} P(s, r) &= \beta_s s - \beta_r r + b_P, \\ Q(s, r) &= \delta_s s + \delta_r r + \tau_{\text{tech}} + b_Q, \end{aligned}$$

so that

$$P(s, r) + Q(s, r) = A_s s + A_r r + B_{\text{tech}}.$$

Let

$$T \equiv \exp\{-[P(s, r) + Q(s, r)]\}, \quad W \equiv \exp\{\gamma P(s, r)\}.$$

The regime's objective can be written as

$$u(s, r) = \bar{V} - \alpha T - \frac{1}{2} k_s s^2 - \alpha c_t W - \frac{1}{2} \phi k_o r^2.$$

Assume an interior solution (s^*, r^*) . The first-order conditions are

$$u_s = \alpha A_s T - k_s s - \alpha c_t \gamma \beta_s W = 0,$$

and

$$u_r = \alpha A_r T + \alpha c_t \gamma \beta_r W - \phi k_o r = 0.$$

Equivalently,

$$k_s s^* = \alpha [A_s T^* - c_t \gamma \beta_s W^*],$$

and

$$\phi k_o r^* = \alpha [A_r T^* + c_t \gamma \beta_r W^*].$$

2.2 Concavity and existence of interior optimum

The Hessian is

$$H = \begin{pmatrix} h_{ss} & h_{sr} \\ h_{sr} & h_{rr} \end{pmatrix},$$

where

$$\begin{aligned} h_{ss} &= -\alpha A_s^2 T - \alpha c_t \gamma^2 \beta_s^2 W - k_s < 0, \\ h_{rr} &= -\alpha A_r^2 T - \alpha c_t \gamma^2 \beta_r^2 W - \phi k_o < 0, \end{aligned}$$

and

$$h_{sr} = -\alpha A_s A_r T + \alpha c_t \gamma^2 \beta_s \beta_r W.$$

To establish strict concavity, consider any nonzero vector $z = (z_s, z_r)'$, we have

$$z' H z = h_{ss} z_s^2 + 2 h_{sr} z_s z_r + h_{rr} z_r^2.$$

Substituting the Hessian terms gives

$$\begin{aligned} z' H z &= (-\alpha A_s^2 T - \alpha c_t \gamma^2 \beta_s^2 W - k_s) z_s^2 \\ &\quad + 2 (-\alpha A_s A_r T + \alpha c_t \gamma^2 \beta_s \beta_r W) z_s z_r \\ &\quad + (-\alpha A_r^2 T - \alpha c_t \gamma^2 \beta_r^2 W - \phi k_o) z_r^2. \end{aligned}$$

Rearranging terms,

$$\begin{aligned} z'Hz &= -\alpha T (A_s^2 z_s^2 + 2A_s A_r z_s z_r + A_r^2 z_r^2) \\ &\quad - \alpha c_t \gamma^2 W (\beta_s^2 z_s^2 - 2\beta_s \beta_r z_s z_r + \beta_r^2 z_r^2) \\ &\quad - k_s z_s^2 - \phi k_o z_r^2. \end{aligned}$$

Equivalently,

$$z'Hz = -\alpha T (A_s z_s + A_r z_r)^2 - \alpha c_t \gamma^2 W (\beta_s z_s - \beta_r z_r)^2 - k_s z_s^2 - \phi k_o z_r^2.$$

Because $T > 0$, $W > 0$, $\alpha > 0$, $c_t > 0$, $\gamma > 0$, $k_s > 0$, $\phi > 0$, and $k_o > 0$, the first two terms are weakly negative and the last two terms are strictly negative for any nonzero vector z . Therefore,

$$z'Hz < 0 \quad \text{for all } z \neq 0.$$

Hence, the Hessian is negative definite globally.

Thus, over a convex feasible set, any interior solution satisfying the first-order conditions is unique and globally optimal.

3 Strategic Substitutability vs. Complementarity

3.1 Substitution vs. Complement Regimes

A key property of the model is that the two strategies, s and r , can be strategic substitutable or complementary, depending on the sign of the cross-partial Hessian h_{sr} :

$$h_{sr} = \alpha(c_t \gamma^2 \beta_s \beta_r W - A_s A_r T).$$

Before examining the two regimes, it is important to clarify the meaning of the key parameter: A_r . When $A_r > 0$, the repression's deterrence effect dominates its identification-undermining effect; when $A_r < 0$, the identification-undermining effect dominates the deterrence effect.

The deterrence effect is likely to dominate the identification-undermining effect ($A_r > 0$) when dissent is broad, public, and low-commitment. For example, ordinary citizens deciding whether to join a protest, sign a petition, attend a meeting, or share oppositional content may be easily deterred by arrests, police violence, job loss, or public punishment. Under these conditions, repression causes many potential participants to withdraw before they become active dissidents. Although repression may make some hard-core dissidents harder to identify, its larger effect is to discourage a broad pool of potential dissidents from participating in politics, thereby reducing threats to the regime. Most modern dictatorships likely fall into this category, as contemporary revolutions often rely less on armed struggle or organized resistance than on crowds, public space, and rapid mobilization (Beissinger, 2022; Bennett and Segerberg, 2023). Protests, strikes, petitions, public criticism, and mass mobilization all require visible participation. Because these forms of dissent depend on citizens' willingness to act publicly, repression can deter them and reduce regime threats, even if it simultaneously makes hidden beliefs or committed dissidents harder to observe.

By contrast, the identification-undermining effect is likely to dominate the deterrence effect ($A_r < 0$) when dissent is narrow, clandestine, and high-commitment. For example, committed dissidents engaged in underground organizing, covert networks, or long-term resistance may be less easily deterred by arrests, police violence, job loss, or public punishment. Under these conditions, repression may not cause core dissidents to withdraw; instead, it may teach them to conceal their activities more effectively, communicate more cautiously, avoid public exposure, and rely on smaller, more trusted networks. In this world, repression reduces the regime's ability to identify dissidents more than it reduces dissidents' willingness to oppose the regime. Although visible punishment may still deter some peripheral supporters, its larger effect is to push committed opponents further underground, thereby weakening the informational benefits of surveillance and making threats harder to detect. Secessionist movements, underground religious groups, armed resistance organizations, and highly repressive contexts such as North Korea likely fall into this category. In these cases, opposition activity depends less on mass public participation than on organized, disciplined, or clandestine forms of resistance. Repression can therefore undermine identification, degrade the quality of information available to the regime, and leave the regime facing a smaller but more hidden and potentially more resilient opposition.

A_r , together with several other parameters, determines the sign of the cross-partial derivative, h_{sr} . Specifically, the sign of h_{sr} depends on whether the workload/enforcement channel, $c_t\gamma^2\beta_s\beta_rW$, dominates the threat-reduction substitution channel, A_sA_rT . The sign of h_{sr} , in turn, determines whether surveillance and visible repression are strategic substitutes or strategic complements.

- Strategic Substitutability ($h_{sr} < 0$): When $A_r > 0$ is large and residual threat reduction is especially important—for example, when baseline identification and deterrence, B , are weak and targeted enforcement c_t is cheap—surveillance and repression become strategic substitutes. In this case, repression reduces threats mainly through deterrence, while surveillance reduces threats through identification and deterrence. Because both instruments lower the same residual-threat term, greater reliance on one reduces the marginal value of the other.
- Strategic Complementarity ($h_{sr} > 0$): When $A_r > 0$ is small, or when $A_r < 0$, and the enforcement workload is costly, the workload externality can dominate. This is especially likely when targeted enforcement c_t is costly, so that a higher level of identification P generates a large burden for the regime. In this case, surveillance and repression become strategic complements through the enforcement channel. Greater surveillance raises P , which increases the number of dissidents the regime must process or punish. Repression, by pushing dissent underground and reducing observable opposition, lowers P and thereby helps contain the enforcement burden created by surveillance. Thus, higher surveillance can increase the regime's incentive to use repression, while repression can make surveillance less costly to sustain.

3.2 Stability of strategic regimes

The distinction between strategic substitutability and strategic complementarity is governed by the sign of

$$h_{sr} = -\alpha A_s A_r T^* + \alpha c_t \gamma^2 \beta_s \beta_r W^*.$$

Equivalently, define

$$\Lambda \equiv \log \left(\frac{c_t \gamma^2 \beta_s \beta_r W^*}{A_s A_r T^*} \right),$$

so that

$$\Lambda > 0 \iff h_{sr} > 0 \iff \text{strategic complementarity},$$

whereas

$$\Lambda < 0 \iff h_{sr} < 0 \iff \text{strategic substitutability}.$$

Hence, the two strategic regimes are separated by the threshold

$$\Lambda = 0,$$

or equivalently,

$$c_t \gamma^2 \beta_s \beta_r W^* = A_s A_r T^*.$$

The regime switches only when a parameter change is large enough to move the economy across this threshold. Therefore, substitutability and complementarity are not continuously or mechanically changing states. They are locally stable as long as the economy is sufficiently far from

$$\Lambda = 0.$$

For a generic parameter θ , a local change shifts the strategic logic according to

$$\frac{\partial \Lambda}{\partial \theta}.$$

Starting from a substitutability region, $\Lambda < 0$, the model switches to complementarity only if

$$\Lambda(\theta_1) > 0.$$

Using a first-order approximation, this requires

$$\theta_1 - \theta_0 > \frac{-\Lambda(\theta_0)}{\frac{\partial \Lambda}{\partial \theta}}, \quad \text{when } \frac{\partial \Lambda}{\partial \theta} > 0.$$

Thus, the farther the economy is from the threshold, the larger the shock required to change the strategic regime.

Similarly, starting from a complementarity region, $\Lambda > 0$, the model switches back to substitutability only if

$$\Lambda(\theta_1) < 0.$$

Locally, this requires

$$\theta_1 - \theta_0 < -\frac{\Lambda(\theta_0)}{\frac{\partial \Lambda}{\partial \theta}}, \quad \text{when } \frac{\partial \Lambda}{\partial \theta} > 0.$$

Thus, the two regimes are relatively stable when

$$|\Lambda|$$

is large. A large positive Λ means the workload channel strongly dominates the residual-threat channel, so complementarity is robust. A large negative Λ means the residual-threat channel strongly dominates the workload channel, so substitutability is robust.

By contrast, regime switching is easy only when

$$|\Lambda| \approx 0.$$

Near this boundary, small changes in α , ϕ , γ , τ_{tech} , K_s , or c_t can change the sign of h_{sr} . In that case, the regime is fragile: surveillance and repression can quickly move from substitutes to complements, or from complements to substitutes.

In substantive terms, substitutability and complementarity are best understood as two locally stable logics of control. A regime does not switch between them simply because one parameter changes slightly. It switches only when the balance between the residual-threat channel and the targeted-enforcement workload channel crosses the threshold

$$\frac{W^*}{T^*} = \frac{A_s A_r}{c_t \gamma^2 \beta_s \beta_r}.$$

3.3 Surveillance Authoritarianism vs. Punitive Surveillance State

Clarifying the substitution and complementarity logics helps explain the theoretical distinction between *surveillance authoritarianism* and the *punitive surveillance state*. The mapping is not perfect, since the model describes the marginal relationship between surveillance and visible repression, whereas the chapter's categories describe broader coercive systems. Still, these model-implied logics provide useful microfoundations for the broader types. The substitution logic helps explain why surveillance authoritarianism is possible: when visible repression undermines the informational value of surveillance or is politically costly, regimes have incentives to rely more on covert surveillance and subtle repression. By contrast, the complementarity logic helps explain why punitive surveillance states emerge: when deterrence dominates or targeted enforcement is costly, surveillance and visible repression can reinforce each other.

To be more specific, under *surveillance authoritarianism*, surveillance and visible repression tend to operate as strategic substitutes. This corresponds to a setting in which the identification function of surveillance remains central, while overt repression is costly or risks pushing dissidents underground. The regime therefore relies more on covert surveillance, targeted enforcement, and subtle punishment than on visible repression. In model terms, it places greater weight on preserving P , the identification effectiveness of surveillance, because visible repression may reduce P , even as it increases deterrence.

Under a *punitive surveillance state*, surveillance and visible repression tend to operate as strategic complements. The regime uses surveillance and overt repression together to deter opposition: surveillance increases citizens' perceived likelihood of detection, while visible repression increases the expected cost of being detected. This logic is especially likely when

targeted enforcement is costly or the pool of potential dissidents is large, making individualized identification and punishment less effective than broad deterrence. However, the mapping is not perfect: surveillance authoritarianism may contain limited complementarity, and punitive surveillance states may contain substitution at the margin.

Thus, both regimes use surveillance for identification and deterrence, but they emphasize these functions differently. Surveillance authoritarianism relies primarily on identification and covert control, whereas punitive surveillance states rely primarily on deterrence and visible coercion. The distinction between substitution and complementarity should be understood as a theoretical tendency rather than a perfect mapping onto regime types.

4 Comparative Statics

Note that closed-form solutions generally do not exist because T and W are exponential functions of the choice variables s and r . In particular,

$$T = \exp\{-(A_s s + A_r r + B_{\text{tech}})\},$$

and

$$W = \exp\{\gamma(\beta_s s - \beta_r r + b_P)\}.$$

Substituting these expressions into the first-order conditions gives

$$k_s s^* = \alpha [A_s \exp\{-(A_s s^* + A_r r^* + B_{\text{tech}})\} - c_t \gamma \beta_s \exp\{\gamma(\beta_s s^* - \beta_r r^* + b_P)\}],$$

and

$$\phi k_o r^* = \alpha [A_r \exp\{-(A_s s^* + A_r r^* + B_{\text{tech}})\} + c_t \gamma \beta_r \exp\{\gamma(\beta_s s^* - \beta_r r^* + b_P)\}].$$

Thus, s^* and r^* appear both linearly on the left-hand side and inside exponential terms on the right-hand side. Because each equation contains a sum of exponential terms involving both choice variables, taking logs cannot isolate either s^* or r^* . The two first-order conditions therefore form a coupled nonlinear system.

For this reason, the model generally cannot be solved in closed form. However, the relationships between the optimal choices, s^* and r^* , and the model parameters can be characterized analytically through comparative statics using the implicit function theorem.

Because the Hessian is negative definite, it is nonsingular. In particular,

$$\Delta \equiv \det(H) = h_{ss}h_{rr} - h_{sr}^2 > 0.$$

Thus, the implicit function theorem applies to the system of first-order conditions

$$u_s(s^*, r^*; \theta) = 0, \quad u_r(s^*, r^*; \theta) = 0.$$

For any parameter θ , the implicit function theorem gives

$$\begin{pmatrix} \frac{\partial s^*}{\partial \theta} \\ \frac{\partial r^*}{\partial \theta} \end{pmatrix} = -H^{-1} \begin{pmatrix} u_{s\theta} \\ u_{r\theta} \end{pmatrix}.$$

Since

$$H^{-1} = \frac{1}{\Delta} \begin{pmatrix} h_{rr} & -h_{sr} \\ -h_{sr} & h_{ss} \end{pmatrix},$$

Thus

$$\frac{\partial s^*}{\partial \theta} = \frac{h_{sr}u_{r\theta} - h_{rr}u_{s\theta}}{\Delta},$$

and

$$\frac{\partial r^*}{\partial \theta} = \frac{h_{sr}u_{s\theta} - h_{ss}u_{r\theta}}{\Delta}.$$

4.1 Cost of visible repression: ϕ

Since

$$u_{s\phi} = 0, \quad u_{r\phi} = -k_or,$$

we obtain

$$\frac{\partial s^*}{\partial \phi} = -\frac{k_or^*h_{sr}}{\Delta},$$

and

$$\frac{\partial r^*}{\partial \phi} = \frac{k_or^*h_{ss}}{\Delta} < 0.$$

Therefore, a higher visible-repression cost always reduces visible repression:

$$\frac{\partial r^*}{\partial \phi} < 0.$$

The effect on surveillance depends on whether surveillance and repression are substitutes or complements. If

$$h_{sr} < 0,$$

then

$$\frac{\partial s^*}{\partial \phi} > 0,$$

so more costly visible repression induces substitution toward surveillance. If

$$h_{sr} > 0,$$

then

$$\frac{\partial s^*}{\partial \phi} < 0,$$

so surveillance and repression move together.

4.2 Share of dissidents: α

Using the first-order conditions,

$$u_{s\alpha} = A_s T - c_t \gamma \beta_s W = \frac{k_s s^*}{\alpha},$$

and

$$u_{r\alpha} = A_r T + c_t \gamma \beta_r W = \frac{\phi k_o r^*}{\alpha}.$$

Therefore,

$$\frac{\partial s^*}{\partial \alpha} = \frac{h_{sr} \phi k_o r^* - h_{rr} k_s s^*}{\alpha \Delta},$$

and

$$\frac{\partial r^*}{\partial \alpha} = \frac{h_{sr} k_s s^* - h_{ss} \phi k_o r^*}{\alpha \Delta}.$$

If $h_{sr} \geq 0$, then both instruments tend to increase with α :

$$\frac{\partial s^*}{\partial \alpha} > 0, \quad \frac{\partial r^*}{\partial \alpha} > 0.$$

If $h_{sr} < 0$, the signs are ambiguous because higher dissident prevalence raises the value of both instruments but also intensifies substitution between them.

4.3 Cost of targeted enforcement: c_t

Since

$$u_{sc_t} = -\alpha \gamma \beta_s W,$$

and

$$u_{rc_t} = \alpha \gamma \beta_r W,$$

we obtain

$$\frac{\partial s^*}{\partial c_t} = \frac{\alpha \gamma W (\beta_r h_{sr} + \beta_s h_{rr})}{\Delta},$$

and

$$\frac{\partial r^*}{\partial c_t} = \frac{\alpha \gamma W (-\beta_s h_{sr} - \beta_r h_{ss})}{\Delta}.$$

When $h_{sr} \leq 0$, these imply

$$\frac{\partial s^*}{\partial c_t} < 0, \quad \frac{\partial r^*}{\partial c_t} > 0.$$

Thus, when targeted enforcement becomes more costly, the regime reduces surveillance because surveillance raises identification and therefore raises targeted-enforcement workload. It compensates by relying more on visible repression, which lowers P through the term $-\beta_r r$.

4.4 Sensitivity to identification: γ

Let

$$L \equiv 1 + \gamma P^*.$$

Then

$$u_{s\gamma} = -\alpha c_t \beta_s W L,$$

and

$$u_{r\gamma} = \alpha c_t \beta_r W L.$$

Therefore,

$$\frac{\partial s^*}{\partial \gamma} = \frac{\alpha c_t W L (\beta_r h_{sr} + \beta_s h_{rr})}{\Delta},$$

and

$$\frac{\partial r^*}{\partial \gamma} = \frac{\alpha c_t W L (-\beta_s h_{sr} - \beta_r h_{ss})}{\Delta}.$$

If $P^* \geq 0$, then $L > 0$. In that case, the comparative statics for γ have the same sign pattern as those for c_t . Under local substitutability, $h_{sr} < 0$,

$$\frac{\partial s^*}{\partial \gamma} < 0, \quad \frac{\partial r^*}{\partial \gamma} > 0.$$

4.5 Technology-specific deterrence: τ_{tech}

Technology enters the residual-threat term through

$$B_{\text{tech}} = b_P + b_Q + \tau_{\text{tech}}.$$

Hence

$$\frac{\partial T^*}{\partial \tau_{\text{tech}}} = -T^*, \quad \frac{\partial W^*}{\partial \tau_{\text{tech}}} = 0.$$

Therefore,

$$u_{s\tau} = -\alpha A_s T^*, \quad u_{r\tau} = -\alpha A_r T^*.$$

Substituting into the implicit-function formula yields

$$\frac{\partial s^*}{\partial \tau_{\text{tech}}} = \frac{\alpha T^* (-h_{sr} A_r + h_{rr} A_s)}{\Delta},$$

and

$$\frac{\partial r^*}{\partial \tau_{\text{tech}}} = \frac{\alpha T^* (-h_{sr} A_s + h_{ss} A_r)}{\Delta}.$$

Since $\alpha > 0$, $T^* > 0$, and $\Delta > 0$, the signs are determined by

$$\text{sign} \left(\frac{\partial s^*}{\partial \tau_{\text{tech}}} \right) = \text{sign} (-h_{sr} A_r + h_{rr} A_s),$$

and

$$\text{sign} \left(\frac{\partial r^*}{\partial \tau_{\text{tech}}} \right) = \text{sign} (-h_{sr} A_s + h_{ss} A_r).$$

If $A_s > 0$, $A_r > 0$, and $h_{sr} \geq 0$, then

$$\frac{\partial s^*}{\partial \tau_{\text{tech}}} < 0, \quad \frac{\partial r^*}{\partial \tau_{\text{tech}}} < 0.$$

Thus, when surveillance and repression are strategic complements, a higher τ_{tech} lowers both optimal surveillance and optimal repression by reducing residual threat.

If $h_{sr} < 0$, the signs are ambiguous because strategic substitution may offset the direct reduction in residual threat.

4.6 Legitimacy Cost of Surveillance: K_s

The parameter K_s is the marginal cost parameter for surveillance. Since

$$u_{sK_s} = -s^*, \quad u_{rK_s} = 0,$$

the implicit-function formula gives

$$\frac{\partial s^*}{\partial K_s} = \frac{h_{rr}s^*}{\Delta} < 0,$$

because $h_{rr} < 0$, $s^* > 0$, and $\Delta > 0$.

For repression,

$$\frac{\partial r^*}{\partial K_s} = -\frac{h_{sr}s^*}{\Delta}.$$

Therefore,

$$\text{sign}\left(\frac{\partial r^*}{\partial K_s}\right) = -\text{sign}(h_{sr}).$$

Hence,

$$\frac{\partial r^*}{\partial K_s} < 0 \quad \text{if } h_{sr} > 0,$$

and

$$\frac{\partial r^*}{\partial K_s} > 0 \quad \text{if } h_{sr} < 0.$$

Thus, a higher cost of surveillance always reduces optimal surveillance:

$$\frac{\partial s^*}{\partial K_s} < 0.$$

Its effect on repression depends on whether surveillance and repression are strategic complements or substitutes. If $h_{sr} > 0$, they are complements, so higher K_s lowers both s^* and r^* . If $h_{sr} < 0$, they are substitutes, so higher K_s lowers s^* but raises r^* .

4.7 Summary

Overall, the comparative statics show that the same shock can lead to different directions depending on the underlying strategic regime. Higher dissident prevalence, lower surveillance costs, stronger technological deterrence, or more expensive targeted enforcement may either intensify both surveillance and repression or shift from one instrument to the other. The key determinant is whether the regime’s control problem is dominated by residual-threat reduction or by the workload costs of identification and targeted enforcement. This distinction provides the model’s main substantive implication: authoritarian control can evolve either toward surveillance authoritarianism, where surveillance substitutes for visible repression, or toward a punitive surveillance state, where surveillance and visible repression reinforce one another.

The distinction between digital and in-person surveillance matters. Digital surveillance is modeled as less costly, meaning it has a lower K_s . This can favor surveillance authoritarianism when surveillance substitutes for repression. Lower surveillance costs allow the regime to expand monitoring while reducing visible coercion. In-person surveillance, by contrast, has stronger deterrent effects, represented by a higher τ_{tech} . This may be more closely associated with punitive surveillance states because in-person surveillance is often experienced as intimidation. It does not merely collect information; it also signals the regime’s coercive presence. Of course, this mapping is not automatic as a technology’s effect depends on whether the regime is in a substitution or complementarity region. Nevertheless, the model implies a general tendency: digital surveillance is more likely to facilitate surveillance authoritarianism because it lowers the cost of covert monitoring, whereas in-person surveillance is more likely to facilitate punitive surveillance states because it combines information gathering with intimidation and deterrence.

The comparative statics for c_t and γ are also important for distinguishing the two regime types. When targeted enforcement is cheap, surveillance authoritarianism is easier to sustain. The regime can identify dissidents and deal with them selectively without needing to rely heavily on public violence. This supports a quieter form of authoritarian control. But when targeted enforcement is costly, surveillance may become burdensome. Identifying more dissidents creates more cases to process, punish, imprison, monitor, blacklist, or otherwise manage. In that setting, the regime may prefer visible repression as a way to deter broad participation rather than individually process every dissident. This pushes the system toward a punitive surveillance state. The model therefore suggests that sustaining surveillance authoritarianism requires not only expanding surveillance capacity, but also lowering the cost of targeted enforcement. This helps explain why regimes may combine mass surveillance with subtler forms of repression, as in China’s social-credit-related mechanisms, which can operate through bureaucratic restrictions and administrative penalties rather than spectacular coercion.

Together, the model shows that the same expansion of surveillance can support two different authoritarian trajectories: a quieter, identification-centered form of surveillance authoritarianism, or a more openly coercive punitive surveillance state.

5 Change between Strategic Substitutability and Complementarity

The strategic relationship between surveillance s and repression r is governed by the cross-partial

$$h_{sr} = -\alpha A_s A_r T^* + \alpha c_t \gamma^2 \beta_s \beta_r W^*.$$

Thus,

$$h_{sr} < 0 \implies s \text{ and } r \text{ are strategic substitutes,}$$

while

$$h_{sr} > 0 \implies s \text{ and } r \text{ are strategic complements.}$$

The cross-partial contains two opposing forces:

$$-\alpha A_s A_r T^*$$

is the *residual-threat channel*. Since both surveillance and repression reduce residual threat, each instrument lowers the marginal benefit of the other. This force pushes toward strategic substitutability.

By contrast,

$$\alpha c_t \gamma^2 \beta_s \beta_r W^*$$

is the *targeted-enforcement workload channel*. Surveillance increases identification and therefore increases the burden of targeted enforcement, while repression reduces this workload. This force pushes toward strategic complementarity.

Therefore, the regime switches from substitutability to complementarity when

$$\alpha c_t \gamma^2 \beta_s \beta_r W^* > \alpha A_s A_r T^*.$$

Since $\alpha > 0$, this is equivalent to

$$c_t \gamma^2 \beta_s \beta_r W^* > A_s A_r T^*.$$

Equivalently,

$$\frac{W^*}{T^*} > \frac{A_s A_r}{c_t \gamma^2 \beta_s \beta_r}.$$

Define the complementarity index

$$\Lambda \equiv \log \left(\frac{c_t \gamma^2 \beta_s \beta_r W^*}{A_s A_r T^*} \right).$$

Then

$$\Lambda > 0 \iff h_{sr} > 0,$$

and

$$\Lambda < 0 \iff h_{sr} < 0.$$

Using

$$T^* = \exp\{-(A_s s^* + A_r r^* + B_{\text{tech}})\},$$

and

$$W^* = \exp\{\gamma(\beta_s s^* - \beta_r r^* + b_P)\},$$

we can write

$$\Lambda = \log\left(\frac{c_t \gamma^2 \beta_s \beta_r}{A_s A_r}\right) + \gamma(\beta_s s^* - \beta_r r^* + b_P) + A_s s^* + A_r r^* + B_{\text{tech}}.$$

Since

$$B_{\text{tech}} = b_P + b_Q + \tau_{\text{tech}},$$

this becomes

$$\Lambda = \log\left(\frac{c_t \gamma^2 \beta_s \beta_r}{A_s A_r}\right) + (A_s + \gamma \beta_s) s^* + (A_r - \gamma \beta_r) r^* + (1 + \gamma) b_P + b_Q + \tau_{\text{tech}}.$$

Hence, the movement toward strategic complementarity is summarized by increases in Λ . The movement toward strategic substitutability is summarized by decreases in Λ .

5.1 Effect of α

α does not by itself change whether surveillance and repression are substitutes or complements, since it scales both the substitutability force and the complementarity force in h_{sr} . The logic shifts toward complementarity only through the way a larger α changes the equilibrium choices (s^*, r^*) . A larger α makes complementarity more likely if

$$\frac{\partial \Lambda}{\partial \alpha} > 0,$$

that is, if

$$(A_s + \gamma \beta_s) \frac{\partial s^*}{\partial \alpha} + (A_r - \gamma \beta_r) \frac{\partial r^*}{\partial \alpha} > 0.$$

Since

$$A_s + \gamma \beta_s > 0,$$

any increase in surveillance induced by a larger α pushes the model toward complementarity. Intuitively, more surveillance raises identification and therefore increases the targeted-enforcement burden W^* . As this burden grows, visible repression becomes more valuable as a tool for containing the costs created by surveillance.

The role of repression is more subtle. Higher repression pushes the model toward complementarity only if

$$A_r > \gamma \beta_r.$$

In that case, repression reduces residual threat T^* more strongly than it reduces the targeted-enforcement burden W^* , so an increase in r^* raises W^*/T^* . If instead

$$A_r < \gamma \beta_r,$$

then higher repression lowers W^*/T^* and pushes the model back toward substitutability.

Substituting the comparative statics for α , we obtain the local condition

$$(A_s + \gamma\beta_s)(h_{sr}\phi K_o r^* - h_{rr}K_s s^*) + (A_r - \gamma\beta_r)(h_{sr}K_s s^* - h_{ss}\phi K_o r^*) > 0.$$

When this condition holds, a larger α raises Λ , moving the regime toward the complementary region.

In other words, because $A_s + \gamma\beta_s > 0$, any increase in surveillance induced by a larger dissident population pushes the system toward complementarity by increasing identification and the targeted-enforcement workload. The repression response is more ambiguous. If $A_r > \gamma\beta_r$, an increase in repression also pushes the system toward complementarity. If $A_r < \gamma\beta_r$, however, an increase in repression pushes back toward substitutability. Even then, the regime moves away from complementarity only if this negative repression effect is large enough to dominate the positive effect of increased surveillance.

Overall, the economic mechanism is

$$\begin{aligned} \alpha \uparrow &\Rightarrow \text{the scale of dissent rises} \Rightarrow s^* \text{ becomes more attractive} \\ &\Rightarrow W^* \text{ rises} \Rightarrow \text{targeted enforcement becomes costly} \\ &\Rightarrow r^* \text{ becomes valuable as a support to surveillance.} \end{aligned}$$

Thus, a larger α makes complementarity more likely when it expands surveillance-intensive control enough that the workload channel dominates the residual-threat channel:

$$c_t \gamma^2 \beta_s \beta_r W^* > A_s A_r T^*.$$

In words, when the opposition threat is large, surveillance alone generates too many costly targets. Visible repression then becomes complementary to surveillance because it helps manage the enforcement burden that surveillance creates.

5.2 Effect of ϕ

The parameter ϕ raises the cost of visible repression. It does not enter the cross-partial

$$h_{sr} = -\alpha A_s A_r T^* + \alpha c_t \gamma^2 \beta_s \beta_r W^*$$

directly. Hence, ϕ affects the strategic relationship between surveillance and repression only indirectly, through its effect on the equilibrium choices (s^*, r^*) .

To characterize this effect, define the complementarity index

$$\Lambda \equiv \log \left(\frac{c_t \gamma^2 \beta_s \beta_r W^*}{A_s A_r T^*} \right),$$

so that

$$\Lambda > 0 \iff h_{sr} > 0.$$

Using

$$\frac{W^*}{T^*} = \exp \{ (A_s + \gamma\beta_s)s^* + (A_r - \gamma\beta_r)r^* + (1 + \gamma)b_P + b_Q + \tau_{\text{tech}} \},$$

we have

$$\frac{\partial \Lambda}{\partial \phi} = (A_s + \gamma\beta_s) \frac{\partial s^*}{\partial \phi} + (A_r - \gamma\beta_r) \frac{\partial r^*}{\partial \phi}.$$

Since

$$\frac{\partial s^*}{\partial \phi} = -\frac{h_{sr}K_or^*}{\Delta}, \quad \frac{\partial r^*}{\partial \phi} = \frac{h_{ss}K_or^*}{\Delta} < 0,$$

it follows that

$$\frac{\partial \Lambda}{\partial \phi} = \frac{K_or^*}{\Delta} [-(A_s + \gamma\beta_s)h_{sr} + (A_r - \gamma\beta_r)h_{ss}].$$

Thus, a larger ϕ moves the regime toward complementarity if and only if

$$-(A_s + \gamma\beta_s)h_{sr} + (A_r - \gamma\beta_r)h_{ss} > 0.$$

Consider first the case in which the regime begins in a substitutability region:

$$h_{sr} < 0.$$

Then a higher ϕ makes visible repression more costly and induces the regime to substitute away from repression and toward surveillance:

$$\phi \uparrow \Rightarrow r^* \downarrow, \quad s^* \uparrow.$$

The increase in s^* raises

$$(A_s + \gamma\beta_s)s^*,$$

and therefore raises W^*/T^* . This strengthens the targeted-enforcement workload channel, making complementarity more likely.

Formally, when $h_{sr} < 0$,

$$\frac{\partial \Lambda}{\partial \phi} > 0$$

if and only if

$$(A_s + \gamma\beta_s)(-h_{sr}) > (A_r - \gamma\beta_r)(-h_{ss}).$$

This condition is especially easy to satisfy when

$$A_r \leq \gamma\beta_r.$$

In that case, the decline in repression also raises W^*/T^* , reinforcing the effect of higher surveillance.

The mechanism is therefore

$$\begin{aligned} \phi \uparrow &\Rightarrow r^* \downarrow, s^* \uparrow \Rightarrow W^*/T^* \uparrow \Rightarrow \text{the workload channel becomes stronger} \\ &\Rightarrow h_{sr} \text{ moves toward positive values.} \end{aligned}$$

However, the effect of ϕ is state-dependent. If the regime begins in a complementarity region,

$$h_{sr} > 0,$$

then higher ϕ lowers both repression and surveillance:

$$\phi \uparrow \Rightarrow r^* \downarrow, \quad s^* \downarrow.$$

The fall in surveillance weakens the targeted-enforcement workload channel and may push the model back toward substitutability, especially when

$$A_r \geq \gamma\beta_r.$$

Thus, the effect of ϕ on the strategic logic is ambiguous in general. Higher repression costs can move the regime toward complementarity when the regime begins in a substitutability region, because costly repression induces greater reliance on surveillance and thereby increases the enforcement burden that makes repression valuable at the margin. But once surveillance and repression are already complements, further increases in ϕ may reduce both instruments and weaken the complementarity logic.

5.3 Effect of τ_{tech}

The parameter τ_{tech} enters directly through

$$B_{\text{tech}} = b_P + b_Q + \tau_{\text{tech}}.$$

Because

$$T^* = \exp\{-(A_s s^* + A_r r^* + B_{\text{tech}})\},$$

a higher τ_{tech} lowers T^* . This weakens the negative residual-threat channel

$$-\alpha A_s A_r T^*.$$

Therefore, holding (s^*, r^*) fixed, higher τ_{tech} pushes the model toward strategic complementarity.

Formally,

$$\frac{\partial \Lambda}{\partial \tau_{\text{tech}}} = 1 + (A_s + \gamma\beta_s) \frac{\partial s^*}{\partial \tau_{\text{tech}}} + (A_r - \gamma\beta_r) \frac{\partial r^*}{\partial \tau_{\text{tech}}}.$$

The direct effect is the +1 term. Therefore, the direct technological-deterrence effect always raises Λ , making complementarity more likely:

$$\tau_{\text{tech}} \uparrow \Rightarrow T^* \downarrow \Rightarrow -\alpha A_s A_r T^* \text{ becomes weaker} \Rightarrow h_{sr} \text{ becomes more positive.}$$

However, τ_{tech} may also reduce equilibrium surveillance and repression. Those indirect effects can partially offset the direct shift toward complementarity. Still, the main logic is:

$$\begin{aligned} \tau_{\text{tech}} \uparrow &\Rightarrow \text{residual-threat substitutability weakens} \Rightarrow \\ &\text{strategic complementarity becomes more likely.} \end{aligned}$$

5.4 Effect of K_s

The parameter K_s raises the cost of surveillance. It does not enter h_{sr} directly, but it lowers equilibrium surveillance:

$$\frac{\partial s^*}{\partial K_s} = \frac{h_{rr}s^*}{\Delta} < 0.$$

Its effect on repression is

$$\frac{\partial r^*}{\partial K_s} = -\frac{h_{sr}s^*}{\Delta}.$$

Thus,

$$\frac{\partial r^*}{\partial K_s} < 0 \quad \text{if } h_{sr} > 0,$$

and

$$\frac{\partial r^*}{\partial K_s} > 0 \quad \text{if } h_{sr} < 0.$$

The effect on the complementarity index is

$$\frac{\partial \Lambda}{\partial K_s} = (A_s + \gamma\beta_s)\frac{\partial s^*}{\partial K_s} + (A_r - \gamma\beta_r)\frac{\partial r^*}{\partial K_s}.$$

Since

$$A_s + \gamma\beta_s > 0$$

and

$$\frac{\partial s^*}{\partial K_s} < 0,$$

the surveillance channel implies

$$K_s \uparrow \implies s^* \downarrow \implies W^*/T^* \downarrow \implies \text{strategic complementarity becomes less likely.}$$

Therefore, higher K_s generally pushes the model toward strategic substitutability by reducing the equilibrium use of surveillance, which weakens the targeted-enforcement workload channel.

The effect through repression is ambiguous. If $h_{sr} > 0$, then higher K_s also lowers r^* , reinforcing the move away from complementarity when $A_r > \gamma\beta_r$. If $h_{sr} < 0$, then higher K_s raises r^* , and this can either offset or reinforce the movement toward substitutability depending on whether

$$A_r > \gamma\beta_r$$

or

$$A_r < \gamma\beta_r.$$

5.5 Effect of Workload Sensitivity: γ

Key insight: workload sensitivity and the emergence of complementarity. The parameter γ governs how strongly surveillance-generated identification translates into targeted-enforcement burden. It therefore plays a central role in determining whether surveillance and visible repression are strategic substitutes or strategic complements.

Recall that

$$h_{sr} = -\alpha A_s A_r T^* + \alpha c_t \gamma^2 \beta_s \beta_r W^*.$$

The first term is the residual-threat channel, which pushes toward substitutability. The second term is the targeted-enforcement workload channel, which pushes toward complementarity. Hence, a larger γ directly strengthens the complementarity force through both γ^2 and W^* .

Define the complementarity index

$$\Lambda \equiv \log \left(\frac{c_t \gamma^2 \beta_s \beta_r W^*}{A_s A_r T^*} \right).$$

Then

$$\Lambda > 0 \iff h_{sr} > 0,$$

so the regime switches to complementarity when

$$c_t \gamma^2 \beta_s \beta_r W^* > A_s A_r T^*.$$

Using

$$W^* = \exp\{\gamma(\beta_s s^* - \beta_r r^* + b_P)\},$$

and defining

$$P^* \equiv \beta_s s^* - \beta_r r^* + b_P,$$

we can write

$$\Lambda = \log \left(\frac{c_t \gamma^2 \beta_s \beta_r}{A_s A_r} \right) + \gamma P^* + A_s s^* + A_r r^* + B_{\text{tech}}.$$

Therefore, holding (s^*, r^*) fixed, the direct effect of γ is

$$\left. \frac{\partial \Lambda}{\partial \gamma} \right|_{s^*, r^*} = \frac{2}{\gamma} + P^*.$$

Thus, a larger γ directly pushes the model toward complementarity whenever

$$P^* > -\frac{2}{\gamma}.$$

Equivalently, if the surveillance-generated identification stock P^* is not too negative, then increasing γ raises the relative strength of the workload channel:

$$\begin{aligned} \gamma \uparrow &\Rightarrow \gamma^2 W^* \uparrow \Rightarrow \text{targeted enforcement becomes more burdensome} \\ &\Rightarrow h_{sr} \text{ becomes more positive.} \end{aligned}$$

Allowing equilibrium behavior to adjust, the full derivative is

$$\frac{\partial \Lambda}{\partial \gamma} = \frac{2}{\gamma} + P^* + (A_s + \gamma \beta_s) \frac{\partial s^*}{\partial \gamma} + (A_r - \gamma \beta_r) \frac{\partial r^*}{\partial \gamma}.$$

Hence, γ makes complementarity more likely if

$$\frac{2}{\gamma} + P^* + (A_s + \gamma \beta_s) \frac{\partial s^*}{\partial \gamma} + (A_r - \gamma \beta_r) \frac{\partial r^*}{\partial \gamma} > 0.$$

The economic logic is straightforward. When γ is low, surveillance primarily reduces residual threat, so surveillance and repression tend to substitute for one another. But when γ is high, surveillance creates a large mass of identified targets. Targeted enforcement then becomes costly, and visible repression becomes valuable because it helps contain that burden. Thus,

$$\begin{aligned} \gamma \uparrow &\Rightarrow \text{surveillance creates more costly targets} \\ &\Rightarrow \text{visible repression supports surveillance} \\ &\Rightarrow \text{complementarity becomes more likely.} \end{aligned}$$

In short, γ is the parameter that converts surveillance from a stand-alone deterrence technology into a burden-generating identification technology. Once that burden is sufficiently large,

$$c_t \gamma^2 \beta_s \beta_r W^* > A_s A_r T^*,$$

visible repression and surveillance become strategic complements.

5.6 Summary

Overall, the model suggests that movement between surveillance authoritarianism and punitive surveillance states is a shift in the strategic relationship between surveillance and repression rather than a change in the level of visible repression. Surveillance authoritarianism arises when surveillance allows the regime to reduce visible coercion and govern through identification and subtle targeted control. It is therefore a particular form of the substitution regime: surveillance is extensive, while visible repression remains relatively limited. Punitive surveillance states arise when surveillance and repression become mutually reinforcing, especially under conditions of costly enforcement, large-scale threats, or technologies with strong deterrent effects.

Surveillance authoritarianism is easier to sustain when targeted enforcement is cheap: the regime can identify dissidents and discipline them through warnings, blacklists, administrative sanctions, mobility restrictions, employment penalties, or other subtle tools. But when c_t or γ rises, surveillance becomes costly precisely because it works: better identification produces more targets to manage. Under those conditions, visible repression becomes more attractive not merely as an alternative to surveillance, but as a support for it. The regime uses repression to deter broad participation and reduce the number of cases that surveillance must convert into individualized enforcement. This is one pathway by which surveillance authoritarianism can transform into a punitive surveillance state.

Importantly, digital surveillance lowers the cost of monitoring and reduces fear, which tends to facilitate surveillance authoritarianism when surveillance substitutes for visible repression. It allows the regime to expand information-gathering while keeping coercion less visible. In-person surveillance, by contrast, often has stronger deterrent and intimidation effects and incurs larger legitimacy costs, making it more likely to support punitive surveillance states. It does not merely identify dissidents; it also communicates coercive presence. Still, the model does not imply a mechanical mapping from technology to regime type. Digital surveillance can be paired with visible repression, and in-person surveillance can be used for targeted identification. The decisive issue is whether the technology strengthens the substitution logic or the complementarity logic.

Note that punitive surveillance states should not be understood as regimes that simply repress more. A regime may face many dissidents or enjoy low repression costs and respond with extensive coercion, but this produces a punitive state, not necessarily a punitive surveillance state. A punitive surveillance state emerges only when surveillance and visible repression become strategically complementary: surveillance makes repression more credible and selective, while repression strengthens the deterrent effect of surveillance. In model terms, this occurs when the targeted-enforcement workload channel dominates the residual-threat substitution channel, so that $h_{sr} > 0$. The key point is therefore not whether repression is high or low, but whether repression operates independently as a blunt tool of domination or jointly with surveillance as part of an integrated system of deterrence and control.

This distinction is crucial for interpreting the effects of dissident scale and repression costs. A larger dissident population, α , does not automatically generate a punitive surveillance state because it increases the overall demand for control rather than directly determining the relationship between surveillance and repression. Similarly, low repression cost does not necessarily produce complementarity. If repression is cheap, the regime can often deter directly through arrests, violence, public punishment, or intimidation without relying heavily on surveillance. Cheap repression may therefore sustain a highly punitive regime, but one based on blunt coercion rather than the systematic fusion of surveillance and repression. It is often when repression is costly, but still useful as a deterrent, that the regime has incentives to combine it with surveillance. Surveillance allows repression to be credible and more threatening, which deters threats without the need to deal with every identified dissident.

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